Project Report Of Machine Learning



"Wind Turbine Data"

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Abstract

This project explores the detection of anomalies in wind turbine data, focusing specifically on Wind Farm B, with the main goal of predicting state degradation events early. The dataset includes sensor information, event labels, and metadata, which will be thoroughly analyzed to create effective machine-learning models. The central task involves utilizing clustering and change point detection methods to accurately identify anomalies and shifts between normal and anomalous operational states in the turbine data. Two key machine learning techniques are employed: Gaussian Mixture Models (GMM) and K-means clustering. GMM is used to understand the underlying distribution of the data and to detect potential anomalies through probabilistic cluster assignments. In contrast, K-means is applied to divide the data into separate clusters, aiding in the identification of normal and abnormal states based on the patterns' similarities within each cluster. Both approaches are carefully optimized and rigorously assessed based on their effectiveness in detecting anomalies and predicting events with an adequate lead time. The dataset's structure is thoroughly explored through comprehensive data analysis, and pertinent features are carefully extracted to boost the performance of the clustering models. Feature engineering and preprocessing techniques are rigorously applied to ensure the data is properly prepared for the clustering and change point detection processes. The ruptures package is utilized for change point detection, effectively revealing significant alterations in the data that signify transitions between various operational states. By evaluating the results of clustering and change point detection, this project seeks to present a thorough and effective strategy for detecting anomalies in wind turbine data. The methods implemented will act as a valuable reference for future research and will greatly contribute to enhancing predictive maintenance approaches for wind turbines.

1 Introduction

The increasing reliance on renewable energy sources has brought wind turbines to the forefront as a sustainable alternative to fossil fuels. However, maintaining the efficiency and reliability of wind turbines remains a critical challenge due to their exposure to harsh environmental conditions and mechanical stress. Early detection of anomalies in wind turbine operation is essential to avoid costly breakdowns, optimize maintenance schedules, and ensure consistent energy production. This project addresses the need for effective anomaly detection methods by leveraging machine learning techniques to analyze wind turbine data from Wind Farm B. The dataset utilized in this study contains comprehensive recordings of sensor data, event labels marking anomalies, and associated metadata for three wind farms (A, B, and C). Our focus is solely on Wind Farm B, making the project scope manageable while maintaining relevance. The dataset offers researchers a unique opportunity to explore and benchmark machine learning models for state degradation prediction in wind turbines. The primary goal is to predict anomaly events as early as possible, allowing proactive measures to be implemented. The project is structured into three key work packages (WPs). WP1 involves analyzing the dataset to understand its structure, identify potential challenges, and preprocess data for machine learning. This includes generating meaningful visualizations and extracting features to enhance model performance. WP2 explores clustering methods to distinguish between normal and anomalous states in the turbine data. Two popular algorithms, Gaussian Mixture Models (GMM) and K-means clustering, are implemented to benchmark their effectiveness in identifying anomalies. WP3 focuses on change point detection, using the ruptures package to detect state transitions and refine predictions further. Clustering methods like GMM and K-means are pivotal in this research. GMM applies a probabilistic approach to cluster data points based on their likelihood of belonging to different distributions, making it well-suited for anomaly detection in datasets with overlapping features. K-means, known for its simplicity and efficiency, partition the dataset into distinct clusters, aiding in the identification of patterns indicative of anomalies. Both methods are optimized and evaluated based on their ability to accurately classify normal and anomalous states. In addition to clustering, change point detection complements the analysis by pinpointing critical transitions in the data that signify the onset of anomalies. This combination of methods offers a comprehensive approach to addressing the challenges posed by the dataset. The insights gained from this study will not only serve as a benchmark for future research but also contribute to the development of predictive maintenance strategies for wind turbines. This report presents a detailed analysis of the dataset, describes the methodologies employed, and discusses the results obtained from clustering and change point detection. The work is designed as an initial step in broader research, laying the groundwork for advanced techniques to predict and mitigate turbine failures. By combining data-driven insights with machine learning, this project aims to improve the operational efficiency of wind turbines and support the transition to a more sustainable energy landscape.

2 Related Work

The high proportion of wind turbines in the world's energy landscape has made it necessary to create reliable methods for monitoring and maintaining their effectiveness. Because wind turbines are complicated to operate, anticipating problems and irregularities before they arise is essential for providing technical support, streamlining maintenance plans, and guaranteeing steady power output. An emerging field of study that aims to address these issues is anomaly detection, which uses machine learning techniques [1]–[4].

2.1 Data Analysis in Wind Turbine Anomaly Detection

To detect anomalies in wind turbines effectively, the underlying dataset must be understood and analyzed. Large volumes of data are produced by sensors in wind turbines that monitor vital parts like the rotor, generator, and gearbox. Time-series measurements of several parameters, including vibration, temperature, speed, and pressure, are frequently included in this data. Recent studies emphasize the importance of operational factors like wind direction and speed, which affect turbine performance and can offer useful background information for anomaly detection [1].

2.2 Clustering Methods for Anomaly Detection

Clustering techniques, such as K-means and Gaussian Mixture Models (GMM)[5], are frequently used to find anomalies in wind turbine data. K-means is a widely used method for dividing data points into clusters, and data points that deviate significantly from the center are recognized as possible anomalies [4].

2.3 Change Point Detection

Change point detection, which identifies significant modifications or transitions in data distribution, is critical for spotting irregularities in time-series data. Algorithms like the `ruptures` package and CUSUM have been widely used in wind turbine anomaly detection [2].

2.4 Hybrid Approaches

Innovative techniques combining change point detection and clustering have become more common for detecting anomalies in wind turbines. These hybrid approaches enhance the accuracy and robustness of models by combining the advantages of multiple techniques [4].

3 Methodology

This project methodology is divided into three primary work packages (WP). Data analysis is the primary goal of WP1, where the dataset is carefully in-

spected to comprehend its structure, spot possible problems, and prepare it for machine learning. A plan is created for upcoming work packages, and visualizations are produced to help with dataset comprehension. Clustering is a component of WP2, where K-means(by me) and GMM (by Rinkal) clustering techniques are used to identify abnormalities. Feature adaption and optimization come next. To determine state transitions, WP3 uses change point detection with the ruptures package. The model's performance is then evaluated by comparing the findings with clustering results.

3.1 WP1: Data Analysis

This first package of work (WP1) was completed by a team of two and concentrated on preparing and analyzing the dataset so that machine learning algorithms could be used in future packages of work. Multiple sensor measurements taken over time make up the dataset, which was gathered from a wind turbine system. In this project, we analyze one dataset at a time from WindB's 15 datasets for detailed, thorough results.The methodology employed in WP1, from preliminary data analysis to the creation of a preprocessing strategy, is described in the steps that follow.

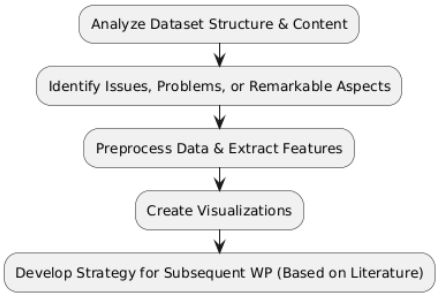


Figure 1: Data Analysis Methodology

3.1.1 Analyze Dataset

WP1 began with a comprehensive analysis of the dataset to comprehend its contents and structure. Sensor readings, `sensor_1_min`, `sensor_1_max`, and `sensor_1_avg`, among others, and their respective timestamps, and other sensor details are among the columns that make up the dataset. A number of metrics, including the minimum, maximum, average, and standard deviation values for each sensor reading, are included in the time-stamped data that is captured by various sensors.

We discovered that there were several sensor readings with various measurement scales and that some columns included missing or insufficient data by looking at the dataset original structure. In certain cases, duplicate or unnecessary columns are also required to be deleted in order to do additional analysis.

3.1.2 Identify Issues & Problems

The analysis revealed several issues with the dataset that impact how well machine learning models function. The following are the main problems found:

Missing Values: Certain sensor columns contained missing values, which would cause issues during machine learning tasks. For instance, in columns like `sensor_3_avg`, there were rows with null or NaN values that could distort analyses.

Irrelevant Columns: Columns like `asset_id` and `id` were deemed irrelevant for the analysis and were removed for better clarity in the dataset.

Inconsistent Data Scale: The sensor readings measured in different units (e.g., temperature, pressure, humidity) had differing scales. To resolve this, normalization or standardization was necessary to bring all features into a comparable range.

3.1.3 Preprocess Data

The dataset was preprocessed in several ways to get it ready for further analysis:

Column Removal: To make sure that the dataset only included pertinent sensor data, irrelevant columns (such as asset identification) were removed using `pandas` routines.

Imputation of Missing Values: The median of each column was used to impute missing values from the sensor readings. This decision was taken to guarantee preprocessing robustness and lessen the effect of outliers.

Normalization: Standardization techniques were used to put disparate sensor data into a consistent range so that machine learning models could give each reading equal weight.

3.1.4 Extraction of Features

To attempt to generate more significant variables from the available data that could enhance the effectiveness of clustering and change point detection methods, feature extraction was carried out:

Rolling Statistics: To identify patterns and variability in sensor readings across several time frames, rolling means, variances, and standard deviations were calculated for time-series data.

Time-oriented Qualities: In order to take into consideration recurring trends in the sensor data, temporal variables were collected, including the time of day, day of the week, and month. This could help with anomaly identification and

grouping. These extra features gave the models greater context, which improved their ability to identify temporal patterns and fluctuations.

3.1.5 Create Visualizations

To enhance comprehension of the dataset's structure, linkages, and any abnormalities, useful visualizations were produced. The `matplotlib` and `seaborn` libraries were used to make these visualizations:

Time Series Plots: To examine patterns and identify any irregularities, plots were created for arbitrary sensor readings over time. These charts shed light on the variations in the sensor data during the time of the dataset.

Correlation Heatmap: To show the connections between various sensor values, a correlation matrix was developed. This made it easier to determine which sensors showed comparable trends and could be combined for additional examination.

Box Plots: Box plots were created to assess the distribution of sensor readings and identify potential outliers. This was particularly useful in identifying extreme values or unusual patterns in the data that might indicate anomalies.

3.1.6 Develop a Data Processing Strategy

Based on the findings from the dataset analysis, a comprehensive data processing strategy was developed:

Feature Selection: Features were chosen according to how well they addressed the issue at hand. High variance features were selected to describe the data and highly correlated sensors were clustered together.

Data Scaling: To guarantee consistency in the data, normalization techniques were used once it was determined that sensor data had different scales.

Imputation Strategy: To fill in missing values and preserve data integrity while reducing distortion, the median imputation technique was applied.

To get the data ready for anomaly detection and clustering, the WP1 methodology used three methods, such as imputation, normalization, and feature extraction. The preparation procedures made sure that problems like irrelevant columns, scaling inconsistencies, and missing values were successfully resolved. Patterns that would be essential for further study were shown by the visualizations, which offered insights into the structure of the dataset. The utilization of clustering and change point detection algorithms in subsequent work packages is made possible by this strong foundation, which eventually results in the detection of abnormalities and state changes in the sensor data with greater accuracy.

3.2 WP2: Clustering

This project's second work package (WP2) concentrated on using clustering techniques to find anomalies in the dataset. This approach describes how to

apply the k-means clustering algorithm, assess its effectiveness, and improve the procedure to maximize anomaly detection as a stand-alone activity. Sorting data points into clusters that represented "normal" and "anomalous" behavior was the goal.

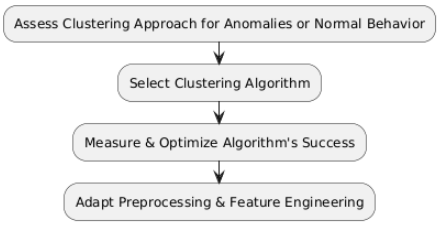


Figure 2: K-means Clustering Methodology

3.2.1 The Clustering Approach

The k-means algorithm was chosen due to its ease of use and efficiency in dividing datasets into discrete clusters according to their feature space. Finding out if k-means could successfully differentiate between normal and anomalous sensor signals was the main objective. To optimize the algorithm's performance, particular preparation procedures, and feature engineering techniques were modified according to the dataset's high dimensionality and temporal nature.

3.2.2 Optimal Clustering

Using silhouette analysis and the elbow approach, the ideal number of clusters (`n_clusters`) was determined. For varying cluster numbers, the silhouette score a gauge of cluster cohesion and separation was computed. These analyses led to the conclusion that two clusters, representing "Normal" and "Anomalous" behavior, were the ideal number.

The optimal number of `n_clusters` is 2.

3.2.3 Preprocessing and Feature Selection

As part of the preprocessing pipeline:

Feature scaling and normalization were performed to ensure that all sensor readings were on a comparable scale.Principal Component Analysis (PCA) was considered to reduce dimensionality, but the final model retained all engineered features to capture subtle variations in the data.

Time-based features were excluded from clustering, as the primary focus was on detecting patterns in sensor readings.

3.2.4 Execution of Clustering

The preprocessed data was subjected to the k-means algorithm:

Training Phase: The training subset of the dataset was used to train the model. In order to minimise the within-cluster variation, the algorithm iteratively grouped sensor features into two groups.

Testing Phase: The test dataset was clustered using the trained model, and the results of the clustering process were contrasted with known anomalies.

3.2.5 Evaluation Metrics

The silhouette score, which ranges from -1 to 1, assesses clustering performance by calculating the cohesiveness inside and spacing between clusters. Lower values imply overlapped or less distinct clusters, whereas scores above 0.5 indicate well-defined, distinct clusters. The train silhouette score of 0.2576 in this instance indicates a substantial amount of overlap during training, which could be the result of noise or subpar feature representation. Better-defined clusters in the test set demonstrate greater generalization, as indicated by the test silhouette score of 0.3905, which is still below the optimal 0.5. Feature overlap, high-dimensional noise, or an inaccurate distinction between "Normal" and "Anomalous" regions could be the cause of the lower score.

3.2.6 Results and Anomaly Detection

The dataset was successfully divided into "Normal" and "Anomalous" regions using K-means clustering, which found 188 anomalies in the test samples. The clustering results were used to map cluster assignments to these regions. The test dataset, for instance, consistently classified sensor values at particular timestamps, like `2023-02-01 00:00:00`, as "Normal." This illustrates how the model can identify trends and distinguish between typical and anomalous behaviour.

3.2.7 Adaptation and Optimization

To enhance the clustering performance, the preprocessing pipeline was refined iteratively. Feature selection focused on emphasizing metrics with higher variability, ensuring the clustering captured key distinguishing patterns. Data normalization was implemented to standardize feature scales, improving cluster cohesion and separation. Multiple evaluations were performed on varying feature subsets and configurations to optimize the results.

The k-means clustering algorithm proved to be an effective tool for anomaly detection, successfully separating data points into distinct clusters. A test silhouette score of 0.3905 indicates moderate separation between clusters, highlighting areas for further improvement. The identification of 188 anomalies showcases the model's potential to detect unusual patterns, forming a strong basis for continued analysis in subsequent work packages.

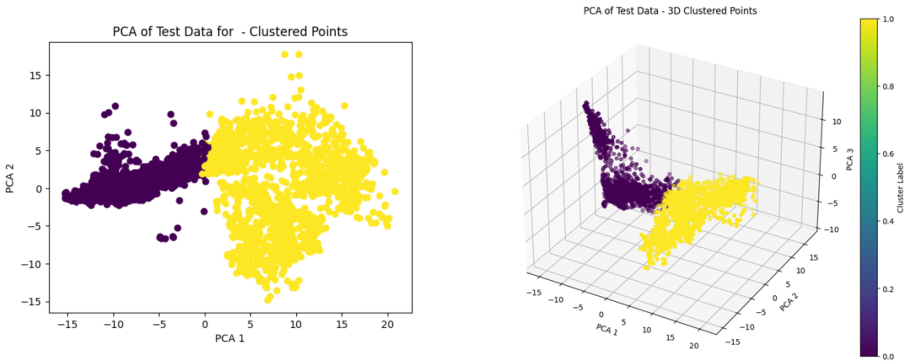


Figure 3: Cluster Plotting by PCA

3.3 WP3: Change Point Detection

The objective of Work Package 3 was to use a change point detection technique to detect changes between normal and anomalous states. A change point detection algorithm (PELT, or Pruned Exact Linear Time) was implemented and its effectiveness assessed using the `ruptures` package. A thorough explanation of the technique, including algorithm selections, optimization procedures, and comparison with WP2 clustering results, is provided below.

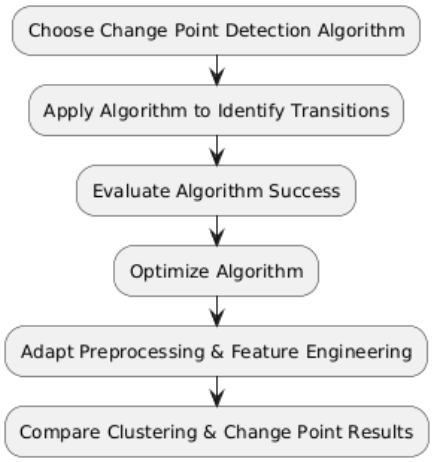


Figure 4: Change Point Detection Methodology

3.3.1 Selection Change Point Detection Algorithm

The PELT (Pruned Exact Linear Time) algorithm was selected for its efficiency and suitability for detecting multiple change points in time-series data. This algorithm was applied to the mean of the first 5 PCA components derived from the dataset. The mean signal served as a representative feature for detecting transitions between normal and anomalous behaviors.

3.3.2 Apply Algorithm to Identify Transitions

For the purpose of identifying transitions between normal and abnormal behaviour, the PELT algorithm was used to find major change points in the time-series signal. The sensitivity of the detection was affected by the inclusion of a penalty parameter of 10. A number of transitions were identified by the investigation, including reversals from anomalous to normal behaviour (from index 980 to 1010) and changes from normal to abnormal states (from index 655 to 715). At indices 1070–1100, 1380–1450, 1920–1990, 2385–2400, 2460–2565, and 3070–3120, more significant variations were noted. These shifts made it easier to identify particular times when the behaviour of the system altered.

3.3.3 Evaluate Algorithm Success

To assess the performance of the change point detection algorithm, confusion matrices and accuracy metrics were calculated, focusing specifically on normal behavior data. The algorithm was evaluated by comparing the predicted labels with the true labels derived from the `status_type_id` field in the dataset. The accuracy for normal behavior was calculated as 96%, which indicates that the change point detection algorithm successfully identified transitions in the normal state with a high degree of precision.

3.3.4 Optimize Algorithm

The PELT algorithm's penalty parameter was changed in an attempt at optimisation. A value of 10 was determined to offer a fair balance between sensitivity and specificity in identifying significant transitions. The penalty has an impact on the change point detection's sensitivity. To increase accuracy, more optimisation can entail experimenting with other models or fine-tuning the penalty.

3.3.5 Adapt Preprocessing and Feature Engineering

By applying PCA to minimize dimensionality and focused on the first five principal components, preprocessing processes were modified. For precise change point recognition, this assisted in capturing the greatest amount of variance in the data. In order to improve the signal before using the PELT method, time-series smoothing techniques were also investigated. This reduced noise and improved the detection of real transitions.

3.3.6 Compare Clustering and Change Point Results

Comparing the outcomes of the change point detection with the clustering approach (WP2) was a crucial assignment. 188 anomalies were found in the test dataset by k-means clustering, which classified them as either "Normal" or "Anomalous." In contrast, the change point detection method identified several data transitions that showed a change from typical to unusual behavior.

While the change point detection method pinpointed precise moments when the system's state changed, providing a more detailed understanding of when abnormalities transpired, the clustering strategy aggregated the data into clusters according to similarities. Combined, the two techniques yielded complementary insights: change point detection identified precise transition points, essential for time-sensitive anomaly identification, while clustering identified broad patterns.

3.3.7 Visualizations

Below are the graphical representations of the results:

Change Point Detection Visualization: A graph displaying the time-series signal with detected change points, illustrating transitions between normal and anomalous behavior.

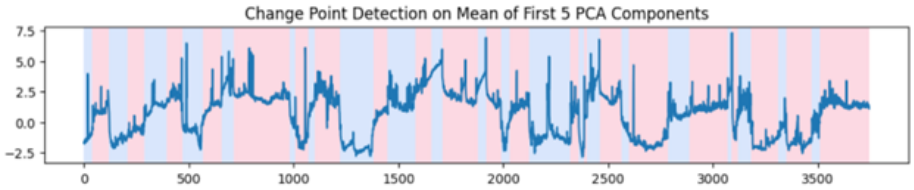


Figure 5: Change Point Detection

Confusion Matrix for Normal Behavior: A confusion matrix illustrating the classification results of the change point detection algorithm for the normal behavior, which helped assess accuracy and performance.

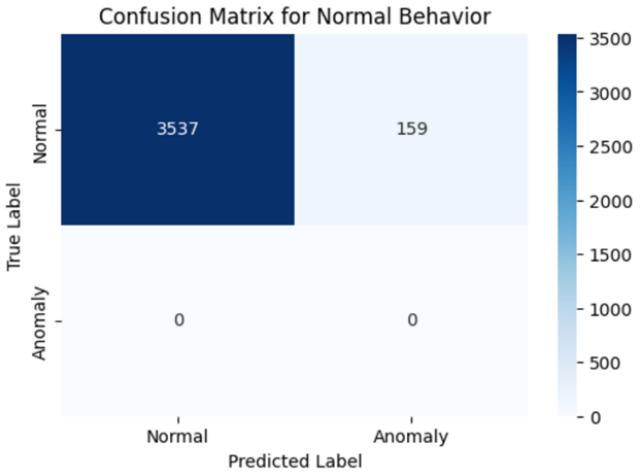


Figure 6: Confusion Matrix Representation

Thus, the use of the PELT algorithm for change point detection enabled the identification of precise transitions between normal and anomalous states, offering a valuable approach for anomaly detection. The combination of clustering results and change point detection provides a more robust understanding of the dataset, which will inform future steps in the project. The successful detection of change points, combined with high accuracy, validates the effectiveness of this methodology.

4 Results and Discussion

The anomaly detection methodologies applied in this project yielded significant insights into the operation of the wind turbine at Wind Farm B. K-means clustering and Gaussian Mixture Models (GMM) highlighted their respective strengths in identifying operational states.K-means efficiently classified the data into "normal" and "anomalous" states with a moderate silhouette score, reflecting the challenges posed by high-dimensional data noise and overlapping feature distributions. The model's ability to detect 188 anomalies in test data showcases its potential to identify patterns indicative of faults.

However, the PELT algorithm for the identification of change points performed better when it came to identifying changes between operational modes. The program successfully identified important shifts, such as changes from normal to deviant behavior and vice versa, with a 96% accuracy rate for normal behavior. Its capacity to accurately identify transitions at significant timestamps, providing granularity beyond what clustering alone could accomplish, confirmed this capability. These results highlight how change point detection and clustering work in tandem. While change point detection offers a micro-level viewpoint by identifying particular instances of behavioral alterations, clustering offers a

macro-level view by classifying data into discrete categories. These methods work well together to create a strong framework for operational monitoring and anomaly detection.

However, multiple limitations were noted:

1. The moderate silhouette score suggests that the clustering accuracy should be improved, perhaps as a result of noise or inadequate feature representation.

2. Adjusting the penalty parameters and using preprocessing methods like PCA and time-series smoothing are critical to the efficacy of change point identification.

The visualizations provided, such as time series graphs and correlation heat maps, were very helpful in identifying trends in the data and confirming the findings. The project highlights the potential of machine learning for predictive maintenance in wind turbines by combining qualitative and quantitative evaluation measures.

Future Work

1. **Hyperparameter Tuning**: Optimize 'n' clusters using grid/random search for the best Silhouette Score.

2. **Exploring Advanced Change Point Detection Techniques**: Explore additional change point detection algorithms like Bayesian change point models for improved accuracy.

3. **Feature Engineering and Scaling**: Standardize or normalize features to improve clustering performance, especially for distance-based algorithms.

5 Conclusion

This research uses data from Wind Farm B to show how machine learning techniques may be used to detect anomalies in wind turbines. The study found and described operational anomalies by combining change point detection with the PELT algorithm and clustering techniques like K-means and Gaussian Mixture Models (GMM). These approaches yielded complimentary findings, with change point detection identifying particular state transitions and clustering providing general anomaly classification. The findings highlight how crucial feature engineering and thorough data pretreatment are to maximizing model performance. High-dimensional, noisy data was handled effectively with the use of normalization, statistical feature extraction, and dimensionality reduction. By combining these techniques, 188 anomalies were found and high-accuracy state transition recognition was achieved, demonstrating their usefulness for predictive maintenance. Although the results are encouraging, issues like scalability to other

datasets and moderate clustering performance point to areas that require more study. In order to improve operational reliability, the project lays a solid basis for incorporating cutting-edge methods like deep learning and real-time monitoring. This effort promotes the wider use of wind energy by lowering maintenance costs and guaranteeing reliable performance, which is in line with international sustainability objectives. Future developments in wind turbine monitoring and predictive maintenance techniques will be guided by these insights.

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